Speed studies

Speeds
- Spot
- Run
- Journey
  - alternate names

Spot speed
- Definition

Running speed
- Defn.
- kaise milega.

Average speed
- Defn

Journey speed
- Defn

Speeds

a vehicle
1. Spot speed
2. Running speed
3. Overall or travel or journey speed
4. Average speed - all vehicles

Spot speed:

it is an instantaneous speed of a vehicle @ specified location.

Running speed

- is the average speed maintained by a vehicle over a particular stretch of road.
- It may be obtained by excluding stoppage or delay

Average speed

is speed of average of spot speed of all vehicles passing a particular point on highway.

Overall speed or travel speed or journey speed

The effective speed with which a vehicle traverses a given route

Speed studies
- Overall speed/ Journey speed
  - kya include krta h.

between 2 terminals
- it is the effective speed of travel which includes stoppage & delays

Present of spot speed
- 98th %ile
  - kya use hz
  - value
- 85th
  - kya specialty hz
  - speed value
- 50th
  - kya hye?
  - alternate name
- 15th
  - %e kischeez ki limit hz
  - value
    - alternate name.

Presentation of spot speed

1) 98th percentile speed
it is used to check highway geometric design.
- 84 kmph

2) 85th percentile speed
it is the upper speed limit or safe speed limit as per Regul?.
- 60 kmph.

3) 50th percentile speed
It is average speed.
. Median speed

4) 15th %ile speed
It is lower speed limit as per regulation.
-30 kmph. [Model speed]

Median speed
- alternate name
- types.
  - TMS
    - Defn
    - formula
  - SMS
    - Defn
    - formula

- kab TMS aur SMS same honge?

- kab traffic volume zero hoga

Median speed (Mean speed)

1 Time meanspeed

It is defined as average speed of all vehicles passing a pt on a highway over specified time.

$$V_t = \frac{\Sigma v_i}{n}$$

2 space meanspeed

is define as avg speed of all vehicles occupying a given section [space] of highway over some specified time period.

$$V_s = \frac{n}{\Sigma\left(\frac{1}{v_i}\right)}$$

Note: Both mean speeds will always be different from eachother except in the unlikely event where all vehicles are travelling @ the same speed.

Note: traffic volume is minimum or zero @ traffic jam or red signal.

~~Fs~~

Traffic density
- Defn
- unit.

Traffic density

The number of vehicles occupying a unit length of a roadway or road @ given instant it is called traffic density. (vehicle/km)

Traffic capacity
- Defn.
- unit.
- formula.
  - formula for S

Traffic capacity

Traffic volume per lane
It is the ability of roadway to accomodate traffic volume per lane.

The capacity is maximum traffic volume or traffic flow.

formula $C = \frac{1000\,V}{S}$

$$\text{Capacity in Vehicle per hour} = \frac{1000 * \text{speed in kmph}}{\text{headway distance or spaceheadway in m.}}$$

$S = (SSD + L)$

↳ length of rigid wheel base b or 6.1.

Headway
- Time headway
  - Defn
  - unit
  - formula
- Space headway
  - Defn
  - formula

Headway

1. Time headway

Gap between two successive vehicles which are crossing a section on road

formula: H_t (in seconds) $= \frac{3600}{\text{Traffic volume in vehicle/hr}}$

$$H_t = \frac{3600}{TV}$$

2. Space headway

H_s = it is the average ~~spe~~ distance between two successive vehicles crossing a particular section

$$H_s = \frac{1}{TD} \quad \text{or} \quad \frac{1}{\text{Traffic density}}$$

$$H_s \text{ (in m)} = \frac{1000}{TD} \rightarrow \frac{\text{vehicle}}{km}$$ ← perfect formula.

7.7

Traffic capacities

- basic capacity
  - alternate names
  - Defn
- Practical cap
  - Defn
  - kyause h2.

Traffic capacities

1. basic capacity or theoretical capacity or ideal capacity

The maximum no of vehicles that pass a given section or a given point on a lane or roadway during 1 hour ideal roadway & traffic condition.

2. Practical capacity (Design cap)

The maximum number of vehicles that can pass a given point so that they may not cause unreasonable delay, hazard accident etc to drivers freedom under prevailing conditions.

Road is designed on the basis of design capacity

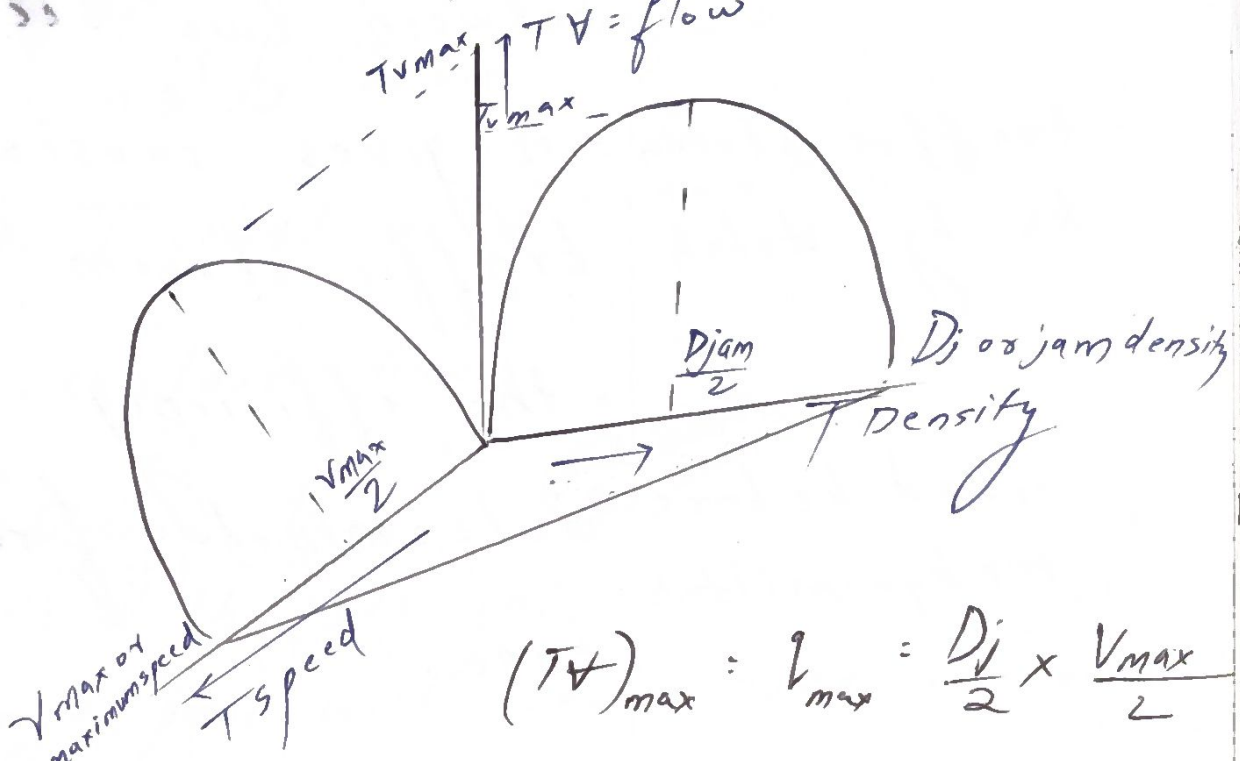
Traffic volume
Traffic density
Traffic speed
relation

- relation formula
- 3D Graph
- formula for maximum traffic volume
- 2D graphs (Greenshields) (Greenberg) (Underwood)

Relation between TV, TD, TS

TV : TD x TS

vehicle / hr : vehicle/km x km/hr

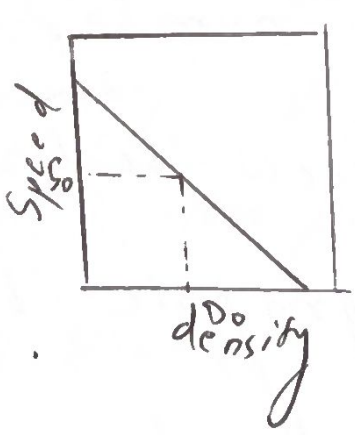


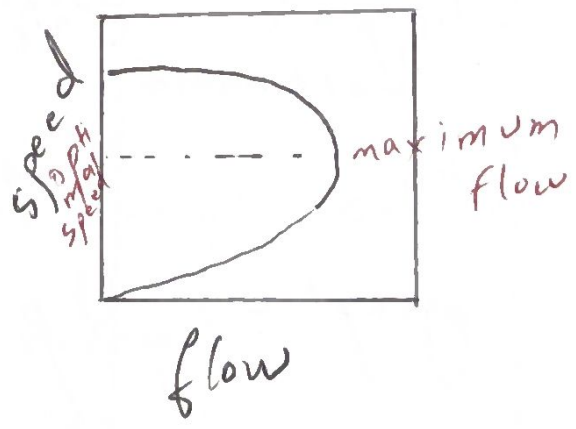
$(TV)_{max} : q_{max} = \frac{D_j}{2} \times \frac{V_{max}}{2}$

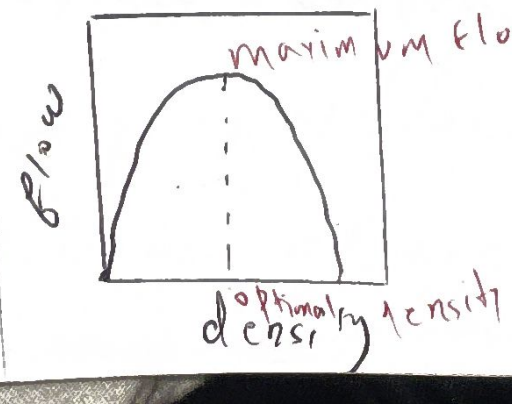
maximum traffic volume.

$q_{max} : \frac{D_j \times V_{max}}{4}$.

max traffic vol = (Jam density x max speed)/4







1.) Speed & delay study.

- kya detah

- traffic stream ka kya detah

- road ke bare me kya milteh

- Methods(5)
  - fcar
    - altname
  - lic plate
    - altname
  - int
  - pgraphic
  - elvtd

Speed & delay study

- it gives information about running speed overall speed fluctuation in speed & delay between two sections.
- it gives causes of delay in traffic stream
- the efficiency of rod is determined from the travel time.

Methods

① floating car method or riding check mthd.

② Licence plate or vehicle number method

③ interview technique

④ photographic technique

⑤ elevated observation.

Origin & destinn studies

- kya dikhata hai?
- Presentn.
  - tab
  - chart
  - contour lines
  - Desire lines imp.
    - kya dikhata hai?
    - width of desire line kya dikhata hai?

Origin and destination studies OD studies

7.10

- This study shows the amount of travel or trip between certain destinations.

- OD study presentn.

1. O & D table
2. π chart or Pie chart.
3. contour lines
4. Desire lines

it is the straight line connecting the origin point with destinations summarised in different area groups.

. the width of desire line is directly proportional to number of trips in both directions

Common methods of both ops speed + delay

- license
- RS interview
- Postcard survey
- sticker/tag
- Home interview
- Workplace interview

Common Methods

1. license plate or vehicle no. methods
2. Roadside interview
3. Postcard survey
4. tag or sticker on vehicle.
5. Home interview.
6. Workplace interview

3 vehicles passing a milestone with 50, 40, 35.3 km per hr speed. What is the time mean speed & space mean speed of vehicles.

Time mean speed

$$V_t = \frac{\Sigma V_n}{n}$$

$n = 3$

$$V_t = \frac{V_1 + V_2 + V_3}{3}$$

$$= \frac{50 + 40 + 35.3}{3}$$

$$= 41.7 \text{ kMPh.}$$

Space mean speed

$$V_s = \frac{n}{\Sigma\left(\frac{1}{V_n}\right)}$$

$$= \frac{3}{1/V_1 + 1/V_2 + 1/V_3}$$

$$= \frac{3}{1/50 + 1/40 + 1/35.3}$$

$$= 40.91 \text{ kmph}$$

Q The design speed of vehicle in traffic lane is 80 kmph Average length of vehicle is 6 m. Then the capacity of vehicle for driver's reaction time 2 second and $f = 0.35$.

Ans $C = \frac{1000 V}{S}$; $V = 80$ kMph given

$S = SSD + L$, $L = 6$ m given

$$= 0.278 V t_R + \frac{V^2}{254 f}$$

$$= 0.278 \times 80 \times 2 + 80^2 / 254 \times 0.35$$

$$= 116.47 \text{ m.}$$

$S = 116.47 + 6 = 122.47$ m

$C = \frac{1000 \times 80}{122.47} = 653.2$ Vehicle/hr

$\simeq 654$ Vehicle/hr

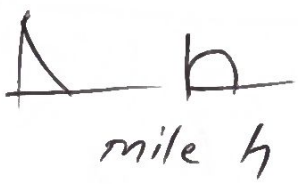
Class 8 12 Aug 24

Traffic stream model
- kisse related h
  - ① G S M S M
    - Aya milta h
  - ② G L M
    - hrg
  - ③ Underwd E M
    - x p

Traffic stream model

- (related to traffic flow/volume) stream

1. Green shield macroscopic model



graphs iss methodce

2. Greenbergs logarithmic method.

3. Underwods exponential method

Traffic flow characteristics
- basic maneuvers kya h?
- weaking kinka sumth
- crossing kya h?

Traffic flow characteristic

Diverging, merging, crossing are the basic maneuvers

Merging & diverging both operations together called weaving.

. Crossing: when traffic crosses eachother @ 90°, it is called crossing operation.

Q-2)

Traffic flow characteristics
- Rotary kya karta h

- Rotary construction or provision of rotary eliminates crossing & operation & encourages weaving operation.

major : crossing
minor : weaving

Number of lanes		Way		
A	B	A → 1 way B → 1 way	A → 1 W B → 2 W	A → 2W B → 2W
2	2	6 major conflicts = 4 minor = 2	11 maj = 7 min = 4	24 maj = 16 min = 8
2	4	10	17	32
4	4	18	25	44

- kiske liye define kye table.

The above conflict points are defined for roads meeting @ 90° or rt L

Traffic islands
- kya hotah traffic island

Traffic islands

- These are raised constructions within the roadway to form channels through which vehicular traffic may be guided.

① Divisional island.
- kya kya hoteh div islands

1) Divisional islands

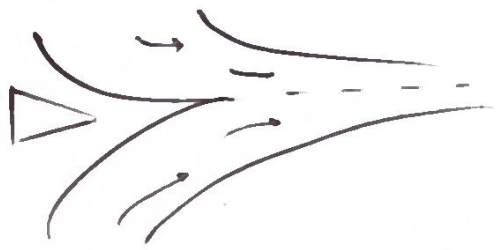
- The kerb/medians/dividers are considered as divisional islands.

② Channelising islands
- use

1) Channelizing islands.

- These are used to guide the traffic into channels at intersection.





③ Ped loading island
- provided @ bus stop
- alt name

3) Pedestrian loading island

- Provided at bus stop also called refuge island

④ Rotary island

4) Rotary island.

8.4)

Traffic islands

- Rotary
  - kya krta h?
  - Rotary Design
    - ① Speed
      - ⓐ Rural roads
      - ⓑ Urban roads
    - ② Traffic vol
      - L Limit lower
      - UL Up limit per
    - ③ Radius
      - formula
        - f_{rural}
        - f_{urban}
        - e

4 Rotary (चौक)

- Rotary or rotary island eliminates (discourages) crossing operation and encourages weaving operation

- Rotary design

1) Design speed

a) rural roads - 40 kmph

b) urban roads - 30 kmph

2) Total volume of vehicles / traffic vol.

lower limit = 500 vehicle / hr

upper limit : 3000 vehicle / hr

3) Radius of rotary

$$R_{min} = \frac{V^2}{127 f}$$

f for rural rods : 0.43

" " urban " : 0.47

e e = 0

Rotary design

④ Width of carriageway
- min

④ Width of carriageway @ entry & exit.

Minimum should be 5 m

⑤ width of weaving sec
- alternate name
- formula
- Range

⑤ Width of rotary roadway or width of weaving section.

$$W = \left(\frac{e_1 + e_2}{2}\right) 13.5$$

(m)
Range of W 6 to 18 m

⑥ length of wvg section
- L formula

⑥ length of weaving section

$$L = 4W$$

⑦ Capacity
- formula Qp
- unit

⑦ Capacity of rotary Qp

formula $Qp = \frac{280 W \left[1 + \left(\frac{e}{W}\right)\right]\left[1 - \frac{p}{3}\right]}{1 + \frac{W}{L}}$

(Vehicle/hr)

- eform

$$e = \frac{e_1 + e_2}{2}$$

- pformula

Weaving ratio $p = \frac{\text{weaving traffic}}{\text{total traffic}} = \left(\frac{b+c}{a+b+c+d}\right)$

e_1 is avg width @ entry
e_2 is avg width @ non weaving section

q.6

Road intersections

- kya hai

Road intersections

- When two or more roads intersect eachother then it is called intersection.

① at grade
- kab hota h

- 1) At grade intersection
  - When roads cross eachother @ same elevation.

② grade separated
- kab hota hai

- 2) grade separated intersection
  - When roads cross eachother @ different level or different elevation.

at grade intersection
- Examples
  - □, skew
  - stag, T, Y

At grade intersection -

square crossing, skew crossing, staggered junction, T jn, Y jn, & multiple intersections.

atgradeisecn

- Types
  - channe.
  - unchan

Types
- Channelised intersection
- unchannelised intersection.

Gradesep intersecn

- alternate name
- flyoverkya h2
- bridge kyah?
- kaunsa crossing traffic ko transfer kr skta h?
- kaunsa highspeed interchange hai?

Grade separated intersection or Interchange

Note: An interchange is also called a fly over

. A brige is an interchange without ramps.

. A diamond crossing is an interchange which provides for transsfer of traffic from one road to another crossing but with reduced speed

. A clover leaf is a high Speed interchange.

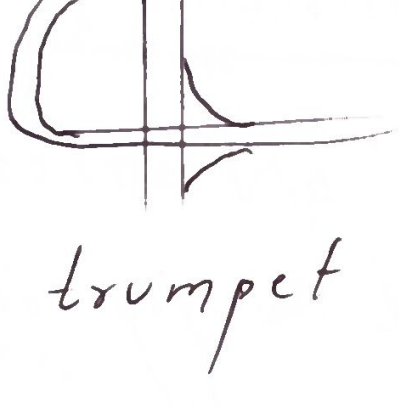
8.30

built up areas me kaunsa interchange lenge aur open areas me kaunsa.

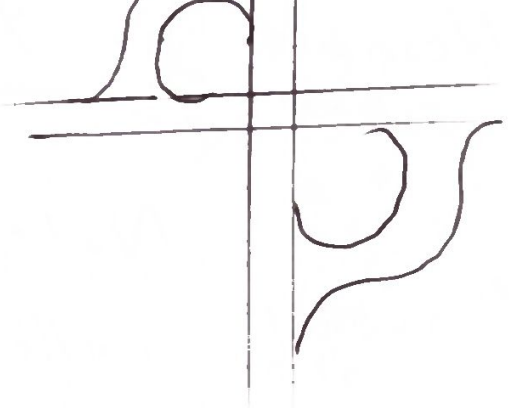
Diamond interchange is used in built up area whereas cloverleaf is preferable in open country or open areas.

Types of interchanges

- tr
- PCL
- SL
- D
- S
- CL



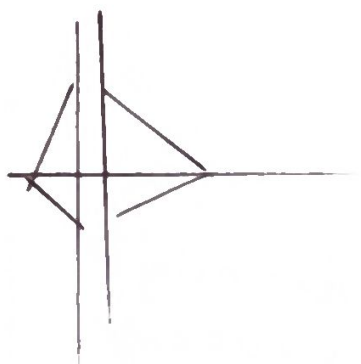
trumpet



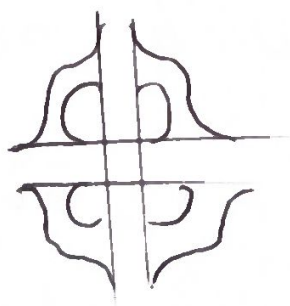
partial cloverleaf



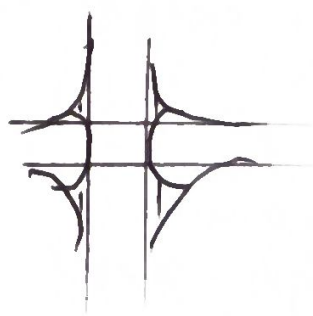
Singlequadrant



Diamond



Cloverleaf



Stack.

Class 9 13 Aug 2024

Parking
- types
  - onstat
  - offstat
- Defns
  - onstreet
  - offstreet
- ways
  - 0° kyah?
    - speciality
  - 90° kyah?
    - kahan use?
    - speciality
  - ∠ parking
    - speciality

Parking:

- on street or on kerb parking
- off street parking

on street parking means vehicles are parked in kerb

off street parking means vehicles are parked in dedicated parking areas.

Vehicles can be parked in -

1) 0° (parallel parking)
- occupies least no of vehicles

2) 90° parking (perpendicular parking)
- adopted when road width is more
- it can occupy max no of vehicles

3) Angle parking 30°, 45°, 60°
- adopted for easy movement of vehicles, more width of road required

Q.2.)

Traffic control devices

- island
- sign
- signal
- markgs

Traffic control devices

1) Trafffic island
2) traffic signs
3) Signal
4) Markings

Q3-1. In desire line diagram width of desire line is a to

no of trips in both directions

Q3-2 A rotary will be more suitab than control by situation listed / against

1) traffic volume entering from all roads is less than 3000 vehicles / hr
3) total right turning traffic is high

The ratio for 30° prking to 60° prking kitnah?

Note
the ratio of width of car parking required @ kerb for 30° to 60° parking = 0.8

3-6) In which of the following grades of a highway is an escape ramp provided?

a) limiting
b) 0 grade
c) down grade
d) Upgrade

Ans down grade

3.7 Capacity of rotary intersection is generally expressed interms of

$$Q_p = \frac{280W\left(1+\frac{e}{W}\right)\left(1+\frac{P}{3}\right)}{1+\frac{W}{L}}$$

① width of weaving section
② length of " section
③ proportion of weaving traffic
④ width of carriageway @ entry

3.8 It is common practice to design highway to accomodate traffic vol corresponding to

30th hour

3.15 Which one of the following is purpose of divisional island.

Ans to segregate opposing flow of traffic in a multilane highway.

3.17 Based on 30th hourly vol how much % of time is the designer willing to tolerate unfavorable conditions

Ans 29.

9.4 Jan '18

journey starts from work to home by walking to the bus travelling by bus to stn & completing the journey is regarded as how many trips

Trip means origin to destination.

1 trip Trip ka matlab origin se destination.

Workplace —walking→ bus —trvl bus→ stn —trvl rail→ Home

origin destinn

1 trip

Q3: two roads two lane two way traffic cross at an uncontrolled intersection number of potential major conflict pts will be

Ans 16 is correct major intersections

Match

Spotspeed	videotape
trafficvol	roadsideint
ODsurve	dopplerradar
Parkingsurve	pneumatic tube

Spotspeed - doppler radar

traffic volume - pneumatic tube

OD survey - road side interview

Parking survey - video tape

Match

Clover leaf interchg	infosign
traffstudies	traffic log dev
rotary/island	Grade sep
Road jn & approach	traf flowchar

Grade separator

traffic flow characteristics

traffic control device

information sign

3.29

Space mean speed	Road condn studies	Correct ans traffic flow study
Journey spd	Regulatory measures	Delaystudy
Running sped	traffic flowstudy	Road condn speed
Spot speed	Resulatory measures	Regulatory measures

3.31 a) Which factor is not strictly related to designd rotary intersechs

1) Radius of central island
2) weaving length
3) Ramps & interchanges
4) Accelern lanes

Ans 3 & 4 are not related rotary de diya toh analyse ramp or accelern lanes lene ki zaroorat nahi h.

modal split — deciding choice for a mode

3.37 Which of the following features requires magnitude of weaving angle and weaving distance for its design a) rotary b) T & L intersecn c) roundabout d) gradesep intersecn

Ans rotary

335 Which set of traffic studies is needed for functional design as well as highway capacity design	Ans speed & volume studies Options a) OD studies b) Parking & accident studies c) Speed & vol study d) Axle load study
335 Which traffic survey scheme is most relevant to deciding major roads in the city	Ans O-D study
Q336 What is used for determining Spot speed	first priority enoscope 2nd radar
337-338 Traffic volume — electronic detector Spot speed — doppler radar Speed & delay — floating car method Multiple character studies — Auto vehicle counter & classifier	

- Traffic signal
  - Provided @?
  - Provided to?
  - consists of 2
  - Signal cycle
    - Defn
  - Types
    - ① Manual
      - Operated by?
    - ② Fixed time
      - kya hyc?
      - kabdelay karwata h?

Traffic signal

- These are provided at road intersections.
- Provided to control, guide, warn the traffic
- it consists of 3 lenses connected vertically → red, amber, green
- Cycle

Defn: The time lapsed from beginning of getting red to a particular side to the beginning of next red to the same side is called a cycle.

Types of road signals.

1) Manual signal

Operated by traffic police

2) fixed time signal

these signals are set to repeat a cycle of sequence & phase of the cycle at regular intervals. it causes delay during off peak hour

9.8)

Trafficsignal
- Types
  - (3) Traffic actuated signal
    - kyakyu?
  - (4) Pedestrian
    - Specialty

3 Traffic actuated signal

- flexible signal cycle as per traffic demand.

4) Pedestrian signal

- it has only two colours red & green

Methods of trafficsignaldesign
- (1) trial cycle method
  - based on?
- (2) Approx method
  - kya purpose?
  - formula
- (3) Websters method

Traffic signal design methods

(1) Trial cycle method

- based on 15 minute count

(2) Approximate method

- minimum green time required for pedestrian traffic

$$G = \left[\frac{W}{1.2} + 7\right] \text{ or } \text{green} = \left[\frac{\text{Width of road}}{1.2 \text{ m/sec}} + 7\right]$$

Paidal chalne wale ki speed.

(3) Websters method

Methods of traffic signal design

- (3) Webstersmthd
  - kaisa method hai?
  - kya nikalte hai?
  - it is based on?
  - formula
    - Co
    - L
      - n is 2 (default)
      - R is ? (default val)
    - y is ? formula

3 Websters method

- This is a national method
- optimum signal cycle time based on least delay at an intersection is worked out
- it is based on saturation flow & normal flow value per unit time on each road meeting at junction

formula for optimum signal cycle time.

$$C_o = \frac{1.5L + 5}{1 - y}$$

L : total lost time

$= 2n + R$.

n is number of phases = 2

R is all red time. = 16 seconds

y is summation of ratio of normal flow to saturation flow traffic for each road $y = y_A + y_B \dots$; $y_A = \frac{N_A}{S_A}$

9.10)

Websters method

- formula
  - Co
  - y
    - Normal flow is determined by 2
    - Saturn flow is?
      - begins with aspect
      - ends with aspect
  - green time
    - road A?
    - road B?

$y = y_A + y_B \ldots$

y_A : Normal traffic a

saturation traffic a

$y_B \ldots$

Normal flow on the road is determined by field studies conducted during peak 15 minute period.

Saturation flow : The maximum continuous flow of one vehicle behind the other vehicle beginning with green aspect or green time and ending with amber or red aspect

Green time -

→ Road A :- $G_A = \frac{y_A}{y}(C_o - L)$

→ Road B :- $G_B = \frac{y_B}{y}(C_o - L)$

Design of traffic signal method

- (4) Is code method
  - based on?

- Red time = ?+?

- Amber time ko green time ke bad kyo detect
  - stop to go
  - stop

(4) Is code mthd

It is based on combination of approximate methods & webster's method

Red time : green time + amber time

Amber time is provided after green time because-

(1) to allow the traffic ~~in~~ change from stop to go.

(2) to allow sufficient time to stop vehicle approaching the intersection beyond SSD.

Traffic signs.
- Types

Traffic signs

- Regulatory or mandatory sign
- informatory sign
- cautionary or warning sign.

12 Traffic signs

- stop sign
  - type
    - shape, colour

Stop sign is a regulatory sign.
with octahedral shape
Red background white text

- Giveway
  - shape
  - type

Giveway △ular sign with apex/x down

it is also a Regulatory sign.

(2) - informatory signs are provided to:
  - marking
  - jn approch

(2) Informatory signs are provided to give information only
- route marking
- road junction approach.

(3) Warning or cautionary sign
  - if not followed then causes

(3) Warning or cautionary sign
- if not followed then causes fatal accident

Class 10 14 Aug 2024

Pavements
- Types
  - flexi
  - rigid

Highway pavement design

Types of pavement

1) flexible
2) rigid

flexible pvmt
- it consits of what at the surface
- it is based on (method of action)
- flexural str of this pvmt is..
- layers
  - S
  - b
  - SG
  - sb

flexible pavement

- it consicts of a series of layers with highest quality material @/or near the surface.
- it is based on function of load distribution through grain to grain action
- flexural strength or bending strength of this pavement is zero or negligible
- it consits of 4 layers

Surface or wearing course
base course
Subbase
Subgrade

5:10-2

Types of roads
hi flexible pvmt

- approx age
- IRC code

Rigid pavement

- Top layer kis kis h
- kasi load transfer kota h?
- kya major factor for design
- kaise failure kojee fleta h?
- kitne layers aur kya kya.
- IRC
- approx life?

- approx age 15 yrs (Range 15)
- IRC 37-2012

Rigid pavement

In rigid pavement top layer is cement/or cement concrete slab

- it can transfer load by slab action.
- its flexural/bending strength is major factor for design.
- it is able to bridge over localised failures.
- It is 3 layered system.

Surface/wearing Course
base
subgrade

- IRC 58:2002
- approx life = 30 yrs & above

flexural str
- ength
  - Definition
  - fcr formula

- flexural ka matlab hota h bending strength
- $f_{cr} = 0.7\sqrt{f_{ck}}$

Semirigid or
- Semiflexible
  - Design factors
    - Design life
    - Defnn
    - Anticipated traffic
    - formula
      - A formula
      - P
      - r
      - n

Semirigid / Semiflexible pavement

Design factors

1) Design life

The number of standard axels that can be carried before strengthening of the pavement is called design life.

2) Anticipated traffic.

$$A = P[1 + r]^n$$

A is Design traffic = no of commercial vehicle / Day
intensity (volume)

P is number of commercial vehicles per day counted on last count

r is growth rate of traffic

n is number of yrs between last count till the end of life of pavement

10.1 Designations of Semi flexible pavement

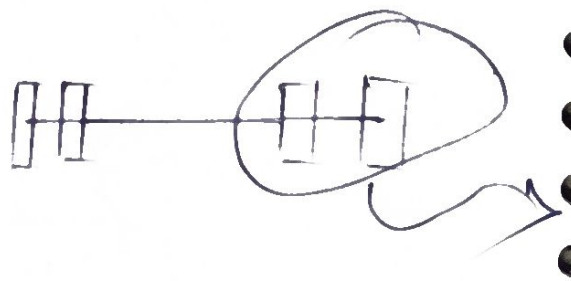
- (3) Design wheel load
  - Defn of max (what is max wl load)
- (4) Equivalent Single wheel load
  - loaded vehicle to maximum wheel load limit ke andr rkhne ke liye kya karna padega
  - diagram
    - distance load refer

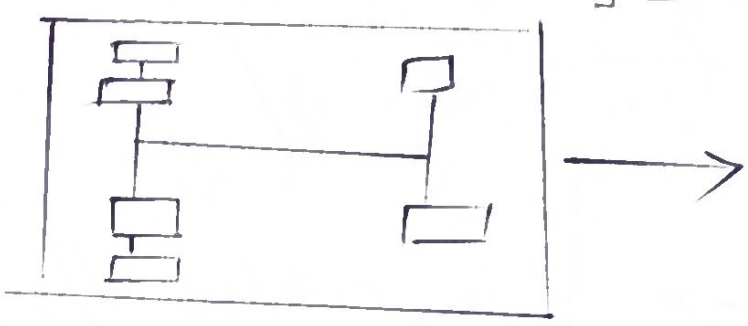
3) Design wheel load (maximum wheel load)

Total load that influences the thickness requirement of pavement is called maximum load

4) Equivalent single wheel load

. To maintain maximum wheel load within the specified limit and to carry a greater load it is necessary to provide dual wheel assembly to the rear axel of load vehicle.

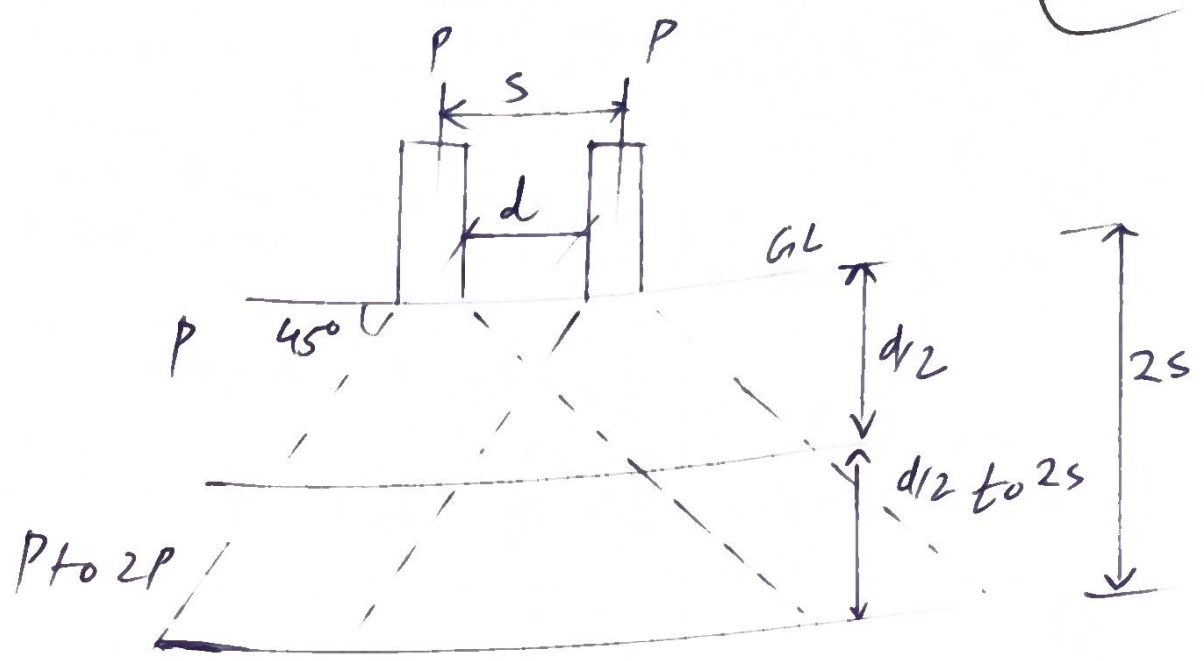




dual wheel rear assembly

ESWL

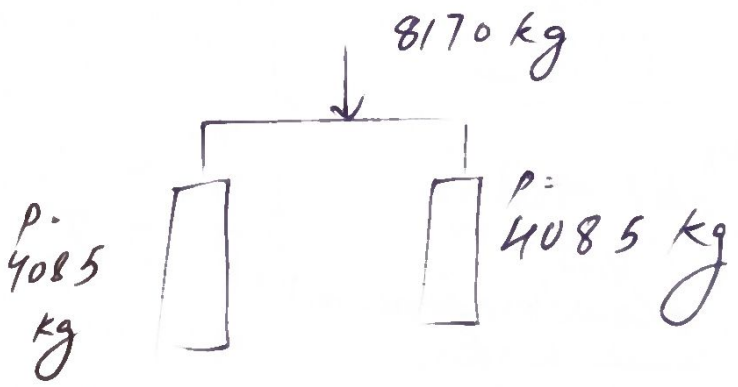
ESWL diagram



- load distance table

distance	load (ESWL)
$d/2$	P
$d/2 < d < 2s$	P to 2P
$d \geq 2S$	2P

- P values



- d kya h?

d: clear distance or inner distance between wheels.

- S kya h?

S: centre to centre spacing between wheels

10 b factors semiflexible pvmt

Design

(5) Contact pressure = ?
- contact area shape?

(6) Tyre pressure
- alternate name

(7) Rigidity factor
- formula
- T or C kiyaye kya hogiwhen
  - RF = 1
  - RF < 1 name
  - RF > 1 name
- kg/cm² ? Min
  - Mpa
  - N/mm²

5) Contact pressure = load on wheel

Contact area is assumed as circle but it is mostly ellipse

6) Tyre pressure = inflation pressure

7) Rigidity factor = $\frac{\text{Contact pressure}}{\text{Tyre pressure}}$

It is the ratio of contact pressure to tyre pressure (RCT)

RF = 1	C · T = 7 kg/cm²
RF < 1 (:: High tyre pressure)	C < 7 kg/cm², T > 7 kg/cm²
RF ? 1	C > 7 kg/cm², T < 7 kg/cm²

(:: low tyre pressure)

0.7 Mpa

0.7 N/mm²

7 kg/cm²

flexible pavement design

Empirical mthd.
- Based on
- Methods(4)
  - G
  - C
  - M
  - S

Semiempirical
- Method

Theoretical
- Method

CBR method
- based on?
- graph

① Empirical methods.

- Based on physical properties or strength of soil subgrade.
- Methods
  - GI, CBR, Mcleod, Stabilometer

② Semiempirical/semitheoretical methods.

- Triaxial method

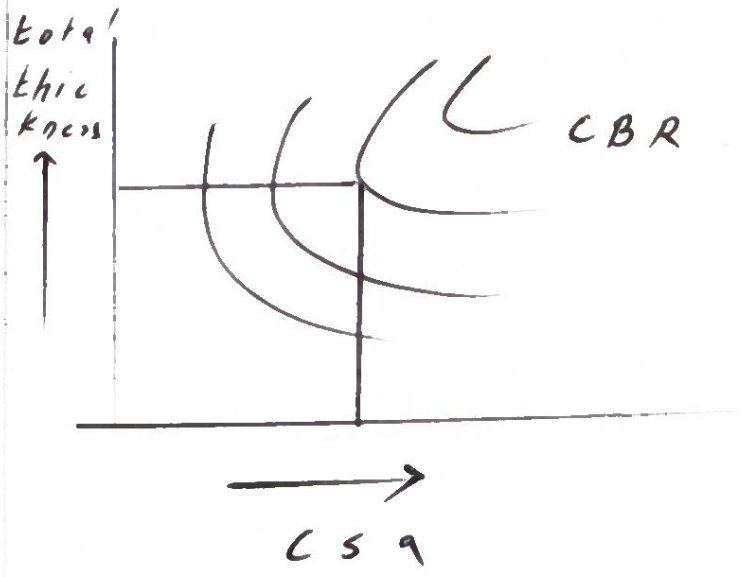
③ Theoretical method

- Burmisters method.

CBR Method

IRC/ recommends method based on cbr &

cumulative standard axle load



10.8

cum standard axle formula?

Cumulative Standard Axle : $\frac{365 \times \left[(1+\text{rate})^{\text{life}} - 1\right]}{\text{rate}} \times$ at present no of commercial vehicles in msa $\times$ lane distrib^{n} factor $\times$ vehicle damage factor

CBR design worksheet me diyah

worksheet 2 - 14 Aug 2024.

minimum. bample size for axle load survey table

* impsecn

minimum sample size for axle load survey	Total number of commercial vehicles per day	% min of commercial vehicles surveyed / sample.
- < 3000	< 3000	20%
- 3000 - 6000	3000 - 6000	15%
- > 6000	> 6000	10%

Class 11 13 Aug 24. 11.1

Equivalency factor
- kya hota hai
- formula

Equivalency factor

it is a factor that converts actual axle load into standard axle load

equivalency factor EF $= \left[\frac{\text{actual axle load}}{\text{standard axle load}}\right]^4$

Triaxial method
- thickness of pavement is kis pe dependent hai our formula.
- E_1, E_2
- T_1, T_2

Triaxial method (semiempirical)

Thickness of pavement required is based on 2 layer systems.

$$\frac{T_1}{T_2} = \left(\frac{Ep_1}{Ep_2}\right)^{1/3}$$

Ep_1, EP_2 are modulus of elasticity of pavement material for layer 1 and layer 2.

T_1 & T_2 are thickness of layer 1 & layer 2

Burmisters mtd
- alternate name

Burmisters method [theoretical]

- layered system

C. Res val method
- based on which two constants
- type of mthd
- formula

California resistance value mthd

- This method is based on two constants:-
  1) Stabilitometer value R
  2) cohesiometer value C
- empirical method
- formula $\left(\frac{T_1}{T_2}\right) = \left(\frac{C_2}{C_1}\right)^{1/5}$

Conclusions
- CBR kya hai
- Max CBR
- GI rank/range
  - Good rank
  - poor rank

Conclusions of flexible pavement design

- CBR is a - measure to indicate relative strength of paving material (imp)
- Maximum value of CBR of any material - 100
- Group index range 0 to 20
  - 0 → Good
  - 20 → poor

Conclusions
- Can GI be -ve

- Group Index cannot be negative if comes out to be -ve, report it 0.

Q Which of the following property is true or correct for soil stabilisation. (Vimp)

Ans (a) subgrade should be treated if it has a soaked CBR 5 or less

(b) base course after treatement Should have plasticity index $I_p \leq 2$. & CBR in excess of 100.

(c) Sub base after treatement should have CBR $I_p \leq 5$

- kab subgrade treat kreinge
- base course CBR I_p after treatment
- Subbase I_p after treatment 2001, 2008

2017 impqn:

GIRL 37 recommendation for CBR method.

ESE 2022

- CBR kis soil pe krteh?
- Soil ko kahan tak compact krteh
- CBR test sample kitne din soak krteh

[Flexible pavemen design]

- CBR test should be performed on Remoulded soil on the field
- for the design of new roads the subgrade soil sample should be compacted @ OMC to prod too density
- for new construction the CBR test sample may be soaked in water for 4 day period before testing